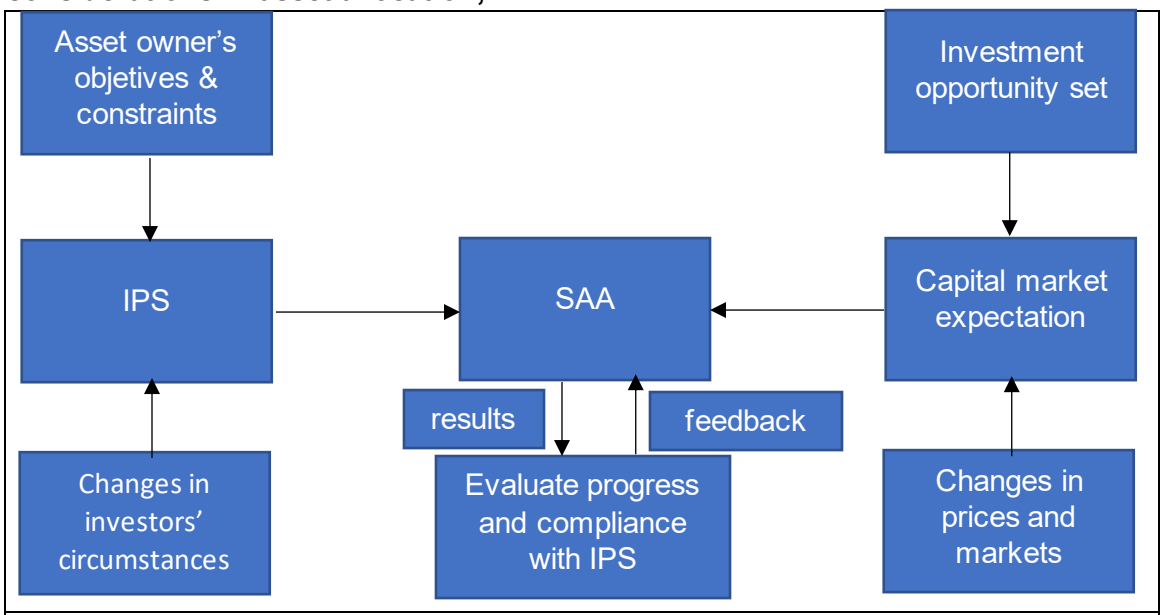
PORTFOLIO MANAGEMENT
LM3: Asset Allocation and Related Decisions in Portfolio Management

LM3: Overview of Asset Allocation

A. describe elements of effective investment governance and investment governance considerations in asset allocation;



INVESTMENT GOVERNANCE – the structure that EXPECTED to ensure that assets are invested to achieve asset owner's investment objectives with risk tolerances and constraints (comply all aplicable rules and regulations)

✓ Investment performance will depend on asset allocation and implementation

- governance structure (institutional investor context) will have 03 levels:

Governing investment committee	BOD members or staff
Investment staff	Large or small in-house staff members
Third-party resources	External investment managers, consultants, custodians, actuaries…

- effective governance has 6 common elements:
  - articulate clear long and short-term objectives of the investment program including:
    - a return objective (numerical) and a verbal objectives (meet future obligation, maintain certain level of liquidity..)
    - risk tolerance: liqudity risk, volatility, risk of loss
  - allocation decision rights and responsibilities among the functional units in the governance hierachy effectively:
    - this should vary depending on size of the staff, knowledge, skills and abilities of in-house staff.
    - delegation to those who are best qualified to make informed decisions.
  - Specify process for developing and approving IPS:

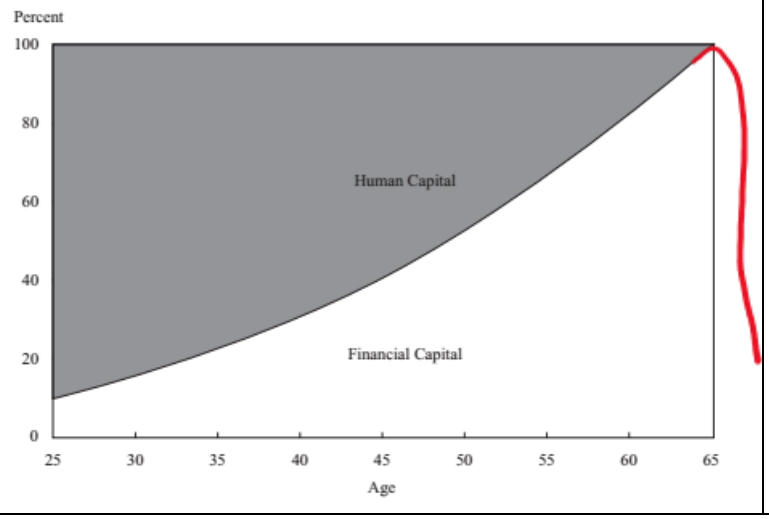
    - Introduction: lay out the purpose and scope of the document and describe the asset owner(s)
    - Statement of investment objectives
    - Investment constraints
    - Duties and responsibilities: investment committee, investment staff and 3rd party providers.
    - Investment guidelines
    - Reporting
- Specify process for developing and approving SAA:
    - Investment committee usually retains approval of SAA decisions
    - Includes information relevant to rebalancing protocol
- Establish a reporting framework to monitor the program's progress towards the agreed-on goals and objectives:
    - It should answer where are we now (absolute sense), where are we relative to the goals and objectives and what value has been generated or deducted by management decisions
    - It should guide: performance evaluation, compliance with investment guidelines and progress towards stated goals
- Periodically conduct a governance audit:
    - Ensure policies, procedures and governance structures are effective
    - Ideally should be conducted by an independent 3rd party

B. prepare an economic balance sheet for a client and interpret its implications for asset allocation;

✓ Economic balance sheet = conventional assets + liabilities + exteded portfolio assets and liabilities such as:
- Human capital (PV of future earnings)
- PV of pension income
- PV of expected inheritances
- PV of future consumption

⇨ Thus asset allocation should take into account the full range of assets and liabilities.

Exhibit 3 Human Capital (HC) and Financial Capital (FC) relative to Total Wealth



Research estimated that human capital is
~ 30% equity like and
~ 70% bond like
(significant variation among industries)

C. compare the investment objectives of asset-only, liability-relative, and goals-based asset allocation approaches;

D. contrast concepts of risk relevant to asset-only, liability-relative, and goals-based asset allocation approaches;

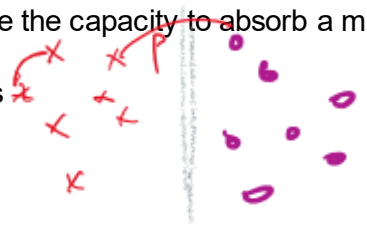
	Asset-only	Liability relative	Goal-based
Definition	Max Sharpe ratio for an acceptable level of risk (Std) and within constraints of IPS	Fund liabilities and invest excess for assets for growth (aslo known as liability driven investing (LDI))	Achieve goals with specified probabilities of success
Characteristics	Focus solely on the asset side of the investors' BS	Choose an asset allocation in relation to the objective of funding liabilities	- Used primarily for individuals and families - Specify asset allocation for sub-portfolios, each of which aligned to specific goals - Each goal is associated with a specific type of cashflow, time horizon and risk tolerance.
Example	MVO - consider only expected return, risk (Std) and correlation of asset classes	Surplus MVO, liability-hedging porfolio + risk porfolio	

Risk metric	- Volatility of return: standard deviation of portfolio - Risk relative to benchmarks: tracking risk - Downside risk: semi-variance, VaR - Monte Carlo simulation: tail risk	- Risk of having insufficient assets to pay obligations when due (shortfall risk) - Volatility of contribution needed to fund current and future liabilities	- Risk of failing to achieve goals: max accetable prob of not achieveing goal - Overall portfolio risk = weighted sum of the risks associated with each goal
Note		Liabilities of institutional investor are LEGAL obligations or debts (unlike GOALS) Liabilities are typically uniform in nature (GOALS are many and varied) Liabilities are numerous and through pooling, law of large numbers and averaging can be forecasted with confidence (GOALS are not subject to the same rules)	

E. explain how asset classes are used to represent exposures to systematic risk and discuss criteria for asset class specifcation;

F. explain the use of risk factors in asset allocation and their relation to traditional asset class–based approaches;

- ✓ Asset class definition (Greer, 1997):
  - Asset of assets that have some fundamental economic similarities to each other
  - Have distinctive characteristics from other assets that are not part of the same class.
- ✓ Criteria for specifying asset classes
  - Asset within a class should be relatively homogenuous
  - Asset classes should be **mutually exclusive**
  - Asset classes should bring diversification benefit (low correlation between different asset classes)
  - Asset class as a group should make up the majority of world investable wealth
  - Asset classes should have the capacity to absorb a meaningful proportion of an investor's portfolio.
- ✓ Specify 3 super classes of assets

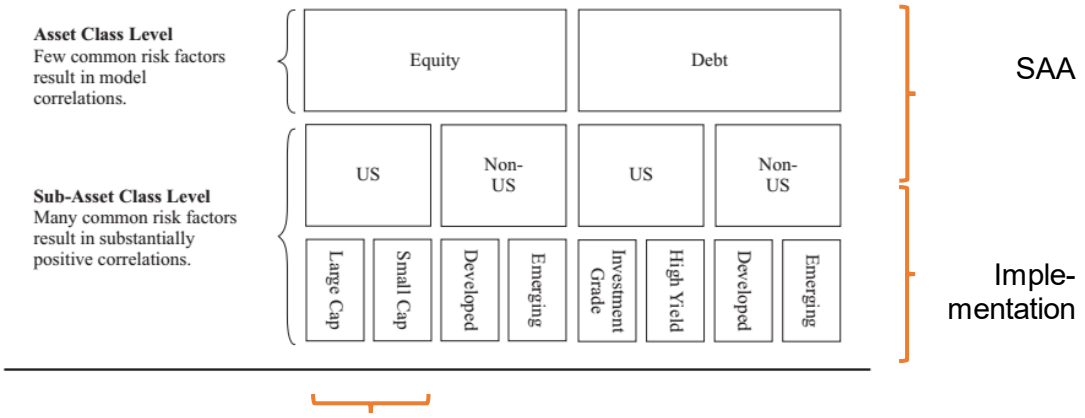


- Capital assets: income producing assets
- Consumable/transformable assets: commodities
- Store of value assets: currencies, gold, art, wine

✓ Asset classes example:
- Global public equity – developed and emerging
- Global private equity – VC/PE
- Global fixed-income – developed and emerging
- Real assets – provide sentivity to inflation
- ⇨ Offer a non-skill-based forecasted expected return premium

✓ Strategies: hedge fund can apply strategies to different asset classes and/or individual investment to generate skill-based return

Exhibit 6 Examples of Asset Classes and Sub-Asset Classes



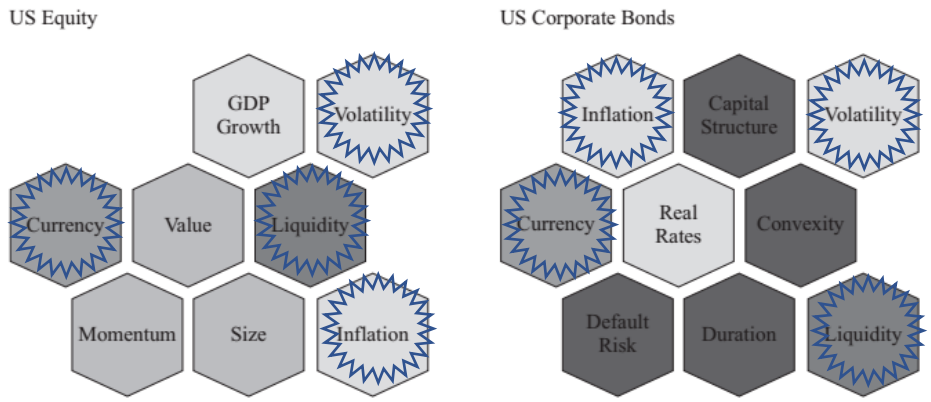
Overlap in sources of risk greater than overlap between US and non-US equity

As more and more sub-asset-classes are defined, they become less distintive

✓ Sources of risk for more broadly defined asset classes are generally better distinguished than those for narrowly defined groups.

✓ Factor-based approaches: use factors instead of asset classes as foundation of portfolio construction:
- Not better but different
- Concentrate on assigning investments for the investors' desired exposure to specific risk factors
- Example

Exhibit 7 Common Factor Exposures across Asset Classes



Overlap in risk factors explain the correlation of equity and bond markets

Asset class allocation obscures the portfolios sensitivity to overlapping risk factors

- ✓ If you use factors as the basis of analysis, please specify risk factors and desired exposure to each. Asset classes could also be described by their sensitivities to each factor.
- ✓ Many highly correlated asset classes are better defined as parts of the same higher asset class

G. select and justify an asset allocation based on an investor's objectives and constraints;

- ✓ Strategic asset allocation = policy portfolio
- ✓ Process of asset allocation:
  - Quantify objectives
  - Establish risk tolerance
  - Determine investment horizon
  - Determine other requirements and constraints
  - Determine suitable asset allocation approach: asset only, liab relative and goal based.
  - Specify asset classes + capital market expection (CME) for each asset class
  - Develop different combination of asset allocation choices for consideration
  - Test/simulate potential stress cases
- ✓ Asset only approach:
  - Porfolios with the most efficient use of asset risk
  - Key metric: Sharpe ratio. A porfolio is efficient if it has the highest Sharpe ratio among portfolios with same volatility of return
  - Sharpe ratio HOWEVER:
    - DOES NOT capture VaR
    - Can not confirm the absolute level of risk is within a specific range
- ✓ Liability relative:
  - Tend to give fixed income a larger role than asset only
  - For underfunded plans, equities would have a higher weight

- Risk factor approaches more relevant factors such as interest rate sensitivity, inflation, credit risk

✓ Goal-based: generally set the strategic asset allocation in a bottom up fashion
- Two different goal classification system:
  - Brunel 2012:
    - Personal goals
    - Dynastic goals
    - Philanthropic goals
  - Chhabra 2005
    - Personal risk bucket
    - Market risk bucket
    - Aspirational risk bucket
- The higher the probability for a future cash need, the greater the amount of assets needed in relation to the need
- Discount rates could be set lower so more assets are assigned to achieve critical sub portfolios (increases the probability of achieving the goal)
- HOWEVER:
  - Sub-portfolios add complexity
  - Goals may be ambiguous or may change over time

H. describe the use of the global market portfolio as a baseline portfolio in asset allocation;

✓ Global market(-value weighted) portfolio
- Sums up all investable assets held by investors
- Reflects the balancing of supply and demand across world markets.

✓ Baseline use:
- Minimize non-diversifiable risk and therefore makes the most efficient use of risk budget
- A reference point for a highly diversified portfolio: investor must provide justification for moving away from global market cap weights
- Provides discipline in mitigating home country bias
- Choice of passive or active investment

I. discuss strategic implementation choices in asset allocation, including passive/active choices and vehicles for implementing passive and active mandates;

✓ Asset class weights
- Tactical aset allocation: deliberate SHORT-TERM deviations from the SAA that reflect short-term views
- Active management at asset class level
- Deviations from SAA usually kept within risk budgets or rebalancing ranges
- Need barriers/monitoring and trading cost (include tax effects)
- Dynamic asset allocation: deviations from SAA motivated by longer-term valuation signals or economic views

✓ Allocation to asset classes
- Passive management: porfolio composition DOES NOT react to changes in CME or to information on invesments

- Active management: will respond to CME and insights/information on investments. Unconstrained active management => benchmark agnostic

✓ Illustration:

MOST PASSIVE ⇨ **Increase tracking risk relative to a benchmark** **MOST ACTIVE**

- Indexing to market weight *(lowest cost approach to investing)*

⇨ **Increase active share relative to a benchmark**

- Unconstrained mandates

Non-cap weighted indexing (NO self-rebalancing)

Traditional, relatively well diversified active strategies

Various aggressive and/or non-diversified strategies

✓ Different asset classes in the same portfolio may be managed with different approaches (active or passive)

J. discuss strategic considerations in rebalancing asset allocations.

✓ Rebalancing means adjusting portfolio weights to more closely align with SAA – due to normal changes in asset prices (other adjustments are not rebalancing)

✓ Rebalancing policy documented in the IPS: normally higher return assets will grow faster than the portfolio => higher return assets also higher risk assets thus a rise in portfolio risk and concentration.

✓ Calendar rebalancing: conduct on a periodic basis (monthly, quarterly, annually)

✓ Percent range rebalalncing: % of portfolio's value around target value act as reference points:

- Example: Target fixed income allocation 50%. So 45-55% could be trigger points.
- May be assessed monthly or quarterly
- Proportional trigger points reflect target weight. Example: 60% asset class => +/- 6%. 20% asset class => +/- 2%
- Rebalancing trade size:
  - Back to target weights
  - Rebalance to range end
  - ½ way between range end and target weight

✓ Strategic consideration:

- Higher transaction cost of an asset class will result in wider rebalancing range
- More risk averse investors > tigher rebalancing ranges
- Belief in momentum => wider ranges
- Illiquid investments are a problem
- Derivatives allow synthetic rebalancing
- Taxes discourage rebalancing, encourage asymetric and wider ranges

✓ Cost – benefit ranges: considers transaction costs and asset class correlation

✓ Illustration:

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Asset class	SAA	Fixed width (10%)	Proportional ranges (+/- 10%)	Cost benefit ranges
Domestic equity	40%	35%-45%	36%-44%	35%-45%
International equity	25%	20%-30%	22.5%-27.5%	19%-31%
Emerging makets	15%	10%-20%	13.5%-16.5%	12%-18%
Fixed income	20%	15%-25%	18%-22%	19%-21%